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MSc Business Analytics

B9BA108 APPLIED RESEARCH PROJECT

Predicting Marine Microplastic Concentration Using Environmental, Spatial, and Anthropogenic Features

A Machine Learning Model Comparison Study

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ABSTRACT

Microplastic pollution has emerged as a major global environmental concern due to the rapid increase in plastic production, inadequate waste management, and the long-term persistence of plastic debris in marine ecosystems. Microplastics, defined as plastic particles smaller than 5 mm, are now detected across all ocean basins, raising serious ecological and human health concerns. Despite growing availability of observational data, predicting spatial variability in marine microplastic concentration remains challenging because of complex oceanographic processes, anthropogenic influences, and heterogeneity in sampling methodologies.

This study develops a machine learning–based regression framework to predict microplastic concentration in ocean waters using an integrated set of environmental, spatial, anthropogenic, and sampling-related variables. A harmonised global dataset was constructed by combining in situ microplastic measurements with oceanographic variables derived from reanalysis data, including ocean current speed, sea surface temperature, and salinity, alongside geographic coordinates, temporal features, population density, and sampling methodology. To address the highly skewed distribution of concentration values, a logarithmic transformation of the target variable was applied.

Five supervised regression models were evaluated: Random Forest, Gradient Boosting, Support Vector Regression, Multilayer Perceptron Neural Networks, and Extreme Gradient Boosting (XGBoost). Following appropriate preprocessing, feature engineering, and hyperparameter optimisation, all models demonstrated substantial improvement in predictive performance. XGBoost achieved the strongest results, explaining approximately 84% of the variance in microplastic concentration on unseen test data ($R^2 \approx 0.84$). Feature importance analysis identified geographic region, ocean current speed, seasonal variability, temperature, and sampling methodology as the dominant predictors.

The findings demonstrate that machine learning, when combined with distribution-aware preprocessing and environmental data integration, offers a powerful and scalable approach for modelling marine microplastic pollution. This work contributes to data-driven marine environmental assessment and provides a foundation for future research integrating time-resolved ocean dynamics and advanced spatio-temporal modelling to support sustainable ocean management.

CHAPTER 1

1.1 INTRODUCTION

The global proliferation of microplastic pollution has been recognised as one of the most urgent and pervasive environmental threats facing marine ecosystems today (UNEP 2021). Microplastics originate from the fragmentation of larger plastic debris, direct industrial release, textile fibres, and consumer product microbeads, contributing to an estimated 8–10 million tonnes of plastic entering the oceans annually (Jambeck et al. 2015). Due to their durability, buoyancy, and resistance to degradation, microplastics can persist for decades, dispersing widely across ocean basins and accumulating in circulation systems such as subtropical gyres (Lebreton et al. 2018). The ecological implications are significant: microplastics can be ingested by plankton, fish, seabirds, and marine mammals, leading to bioaccumulation, physiological stress, and potential human health impacts through seafood consumption (Rochman et al. 2013; Sharma & Chatterjee 2017).

Understanding and predicting the spatial distribution of microplastics is essential for effective environmental management, targeted cleanup interventions, and the development of early warning systems. However, microplastic dispersal is influenced by a complex interplay of factors including ocean circulation, wind-driven transport, seasonal monsoon systems, salinity gradients, vertical mixing, and human population density (Van Sebille et al. 2020). Traditional statistical approaches have struggled to model these non-linear relationships at global scale, limiting the ability of policymakers and environmental organisations to anticipate pollution hotspots or understand the underlying drivers of contamination.

Recent advances in machine learning provide new opportunities for modelling large-scale environmental systems by enabling algorithms to learn complex, multi-dimensional relationships from heterogeneous datasets (Reichstein et al. 2019). Machine learning models have previously demonstrated success in applications such as air quality prediction, hydrological forecasting, and marine habitat classification (Hengl et al. 2018). However, the application of machine learning to microplastic prediction remains limited, with most prior work focusing on ocean drift models, particle tracking simulations, or region-specific assessments (Li et al. 2021; Maximenko et al. 2019). There is a clear need for research that integrates environmental, spatial, and anthropogenic data into predictive frameworks capable of generalising across diverse marine contexts.

This dissertation addresses this gap by evaluating the effectiveness of multiple machine learning algorithms—Random Forest, Gradient Boosting, Support Vector Regression, Multilayer Perceptron, and XGBoost—for predicting microplastic concentration based on integrated environmental and socio-spatial features. Through systematic model comparison, feature importance analysis, and critical evaluation of predictive limitations, the study aims to advance the current understanding of microplastic distribution modelling and contribute to the development of scalable, data-driven forecasting tools that can support sustainable ocean management strategies.

1.2 AIMS AND OBJECTIVES

Aim

To develop and evaluate a machine learning framework capable of predicting microplastic concentration in ocean waters using integrated environmental, spatial, anthropogenic, and sampling-related variables.

Objectives

1. To construct a comprehensive, harmonised dataset
by integrating global microplastic observation records with oceanographic variables (ocean current velocity, temperature, salinity), spatial coordinates, and anthropogenic indicators such as population density (García et al. 2021).
2. To preprocess and engineer features
using appropriate scaling, encoding, and transformation techniques to ensure reliable and consistent input data for machine learning models.
3. To develop and train multiple machine learning algorithms
including Random Forest, Gradient Boosting, Support Vector Regression, Multilayer Perceptron, and XGBoost, using the harmonised dataset.
4. To evaluate the predictive performance
of each model using RMSE, MAE, and R^2 metrics, and determine which algorithm provides the highest predictive accuracy and generalisation capability.
5. To analyse feature importance
with a particular focus on XGBoost, to identify the most influential environmental, spatial, and anthropogenic factors contributing to microplastic concentration predictions.
6. To interpret results within the context of marine environmental science
by examining how ocean dynamics, geographic patterns, and human activities influence model performance and pollution distribution.
7. To identify limitations and develop recommendations
for future work, including integrating time-series hydrodynamic drift data, satellite-derived ocean circulation, and Lagrangian particle tracking models to enhance predictive capability.

1.3 RESEARCH QUESTIONS

Based on the existing gaps identified in marine microplastic prediction research, the following research questions guide this dissertation:

RQ1: To what extent can environmental, spatial, anthropogenic, and sampling-related variables predict microplastic concentration in ocean waters using machine learning models?
This question addresses the core problem of whether static environmental and socio-spatial data can meaningfully explain variability in observed microplastic measurements, as suggested by Van Sebille et al. (2020) and Li et al. (2021).

RQ2: Which machine learning model demonstrates the highest predictive performance for microplastic concentration forecasting?
The goal is to determine whether advanced non-linear models such as XGBoost outperform

traditional models, consistent with findings by Reichstein et al. (2019) regarding non-linear environmental systems.
RQ3: Which environmental and anthropogenic factors contribute most significantly to microplastic concentration prediction according to feature importance analysis?
This evaluates whether features such as ocean current velocity, temperature, geographic region, and population density align with theoretical and empirical research on plastic transport dynamics (Lebreton et al. 2018; Maximenko et al. 2019).
RQ4: What limitations arise when using static environmental variables to predict microplastic concentrations, and how can future models incorporate temporal ocean drift data for improved accuracy?
This addresses the known gap that microplastic movement is inherently time-dependent (Cózar et al. 2014).

Flowchart: Structure of the Report

Structure of the Dissertation

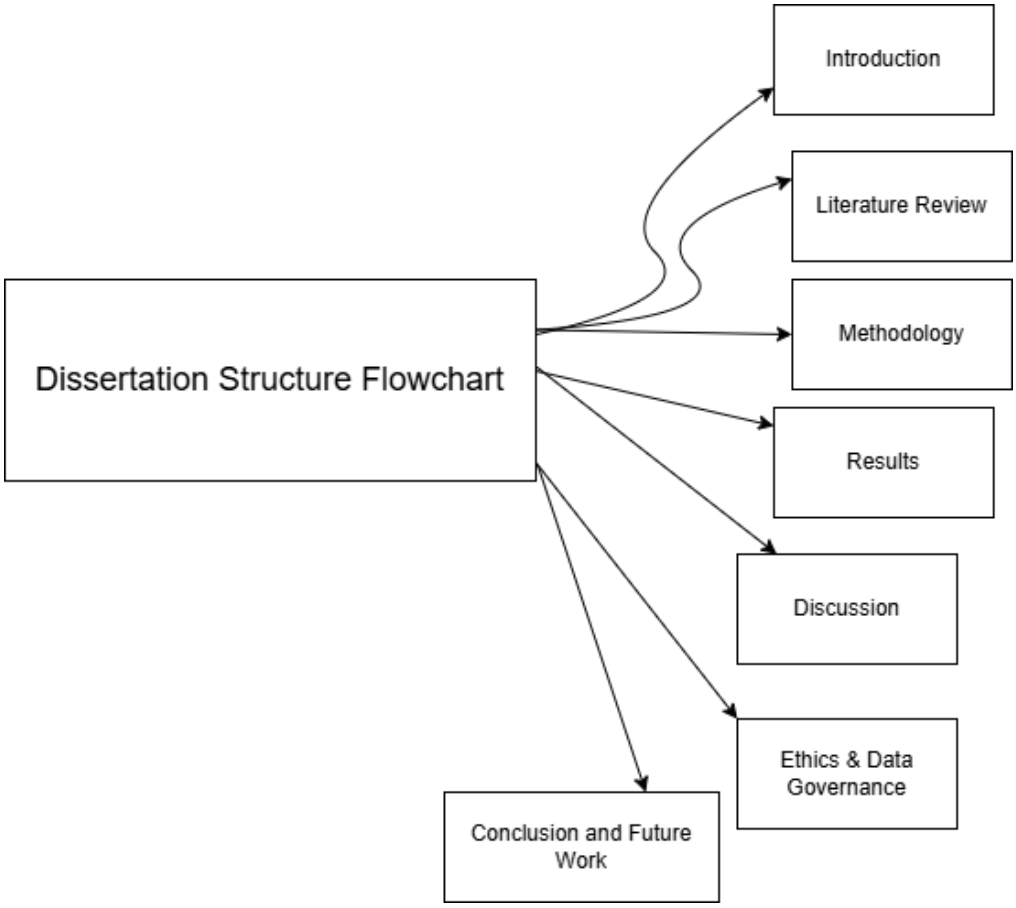


Figure 1: Structure Flowchart

To guide the reader through the scope, methodology, and contributions of this research, the dissertation is organised into a series of interrelated chapters, as illustrated in figure. Each

chapter builds progressively from theoretical foundations to empirical analysis and interpretation.

Chapter-by-Chapter Overview

Chapter 1 – Introduction
Introduces the global challenge of microplastic pollution, establishes the research motivation, defines the problem context, and outlines the aims, objectives, and overall structure of the study.

Chapter 2 – Literature Review
Critically reviews existing research on microplastic pollution, oceanographic transport mechanisms, and the application of machine learning in marine environmental modelling. This chapter identifies methodological gaps that motivate the current study.

Chapter 3 – Methodology
Details the data sources, preprocessing steps, feature engineering strategy, model selection rationale, and evaluation metrics. The machine learning workflow and experimental design are explained in a transparent and reproducible manner.

Chapter 4 – Results
Presents the empirical findings of the study, including model performance comparisons, error metrics, and feature importance analysis. Results are reported objectively without interpretation.

Chapter 5 – Discussion
Interprets the results in relation to the research objectives and existing literature. The implications of model performance, predictor relevance, and observed limitations are critically examined.

Chapter 6 – Ethics and Data Governance
Addresses ethical considerations, data provenance, licensing, bias, and responsible use of environmental and population datasets.

Chapter 7 – Conclusion and Future Work
Summarises key findings, highlights the study's contributions, and outlines recommendations for future research and policy-oriented applications.

CHAPTER 2

LITERATURE REVIEW

2.1. Overview of Microplastic Pollution

Marine microplastic pollution has emerged as one of the most pressing and complex environmental challenges of the twenty-first century. The rapid expansion of plastic production since the mid-20th century, combined with widespread inefficiencies in global waste management systems, has resulted in unprecedented levels of plastic debris entering natural ecosystems. Microplastics are commonly defined as plastic particles with a diameter of less than 5 mm and are categorised into two broad types: primary microplastics, which are intentionally manufactured at small sizes, and secondary microplastics, which result from the fragmentation of larger plastic items through physical, chemical, and biological processes (Andrady, 2011).

The durability, lightweight nature, and resistance to degradation that make plastics commercially valuable also contribute to their long-term persistence in marine environments. Unlike organic pollutants, plastics do not readily biodegrade but instead fragment into progressively smaller particles, increasing their abundance and environmental availability over time. Geyer, Jambeck and Law (2017) estimated that global plastic production has exceeded 8.3 billion tonnes, with a significant fraction entering landfills or the natural environment. Once plastics enter marine systems, recovery becomes practically infeasible, particularly for microplastics due to their microscopic size and extensive spatial dispersion. Empirical studies have documented the presence of microplastics across all major ocean basins, including surface waters, deep-sea sediments, polar regions, and within a wide range of marine organisms (Barnes et al., 2009; Cózar et al., 2014). This pervasive distribution underscores the global scale of the problem and highlights the urgent need for robust analytical and predictive frameworks capable of capturing spatial and temporal variability in microplastic concentrations. Importantly, the heterogeneity of microplastic datasets—arising from differences in sampling methods, spatial coverage, and temporal resolution—poses significant challenges for traditional modelling approaches.

Beyond environmental contamination, microplastic pollution raises serious ecological and human health concerns. Microplastics can act as vectors for persistent organic pollutants, heavy metals, and pathogenic microorganisms, facilitating their transport through marine food webs (Rochman et al., 2013). The ingestion of microplastics by marine organisms has been widely documented, with evidence of bioaccumulation and trophic transfer across multiple ecological levels. These risks have elevated microplastic pollution from a purely environmental issue to one with broader socio-economic and public health implications.

2.2. Sources of Microplastics and Human Influence

Microplastics enter marine environments through a variety of pathways, reflecting both direct emissions and indirect degradation processes. Primary microplastics include cosmetic microbeads, industrial abrasives, and pre-production plastic pellets, which are deliberately manufactured at microscopic sizes for specific applications (Napper and Thompson, 2016). Although regulatory measures have reduced microbead usage in several countries, primary

microplastics continue to contribute to marine pollution through industrial loss and textile fibre shedding during domestic washing.

Secondary microplastics originate from the fragmentation of larger plastic debris such as bottles, packaging materials, fishing gear, and maritime waste. Exposure to ultraviolet radiation, mechanical abrasion, and thermal stress accelerates the breakdown of macroplastics into smaller particles over time (Andrady, 2011). These secondary sources dominate global microplastic abundance, reflecting decades of accumulated plastic waste and ongoing leakage into marine systems.

Land-based sources account for the majority of plastic inputs into the oceans. Rivers serve as major transport pathways, conveying mismanaged plastic waste from inland regions to coastal waters. Jambeck et al. (2015) estimated that between 4.8 and 12.7 million tonnes of plastic waste enter the ocean annually from coastal populations alone. Subsequent studies have refined these estimates, demonstrating that a relatively small number of highly polluted rivers contribute disproportionately to global plastic emissions (Lebreton et al., 2017; Meijer et al., 2021).

Anthropogenic drivers such as population density, urbanisation, industrial activity, tourism, and waste management infrastructure play a critical role in determining plastic leakage rates. Regions characterised by rapid population growth and inadequate waste treatment facilities—particularly in parts of Southeast Asia and South Asia—have been identified as major contributors to marine plastic pollution (Jambeck et al., 2015; Lebreton and Andrady, 2019). These findings highlight the importance of incorporating anthropogenic variables, such as population density, into predictive models of microplastic concentration.

Importantly, human influence extends beyond source generation. Coastal development, port activity, and maritime transport can influence local hydrodynamics and debris retention, further complicating spatial distribution patterns. As such, microplastic pollution must be understood as a coupled human–natural system, requiring integrative analytical approaches that capture both environmental and socio-economic dimensions.

2.3. Environmental Drivers of Microplastic Transport and Accumulation

Once microplastics enter the marine environment, their distribution is governed by a complex interplay of physical oceanographic processes. Surface currents, wind-driven circulation, mesoscale eddies, and large-scale gyres collectively shape the horizontal transport of floating plastic debris across ocean basins (Van Sebille et al., 2020). These processes enable microplastics to travel thousands of kilometres from their original source regions, contributing to their global dispersion.

Subtropical gyres, such as those in the North Pacific and North Atlantic, are well-documented accumulation zones where converging currents concentrate floating debris (Lebreton et al., 2018). However, accumulation is not restricted to surface waters. Vertical transport mechanisms—including biofouling, aggregation with organic matter, and changes in particle density—can cause microplastics to sink or remain suspended at depth (Kooi et al., 2017). Consequently, microplastics have been detected in deep-sea sediments and benthic ecosystems, challenging earlier assumptions that pollution is confined to surface layers.

Environmental variables such as sea surface temperature and salinity influence water density, stratification, and mixing processes, which in turn affect the buoyancy and vertical distribution of microplastics. Warmer surface waters can enhance stratification, potentially trapping particles near the surface, while salinity gradients influence vertical mixing and transport pathways (Van Sebille et al., 2020). These factors provide a strong physical rationale for including temperature and salinity as predictors in data-driven modelling frameworks.

Seasonal variability further complicates microplastic distribution patterns. Monsoon systems, storm events, and seasonal current reversals can significantly alter transport pathways and concentration levels, particularly in coastal and semi-enclosed seas (Isobe et al., 2021). Temporal descriptors such as month and year therefore offer valuable contextual information that can improve model performance by capturing seasonal and interannual variability.

2.4. Spatial Distribution Patterns and Regional Hotspots

Global observational studies reveal pronounced spatial heterogeneity in microplastic concentration. Coastal regions generally exhibit higher concentrations than open ocean areas due to their proximity to land-based sources and reduced dispersal capacity (Cózar et al., 2014). Enclosed and semi-enclosed seas, including the Mediterranean Sea, South China Sea, and Gulf of Thailand, have been identified as persistent pollution hotspots owing to limited water exchange, high population density, and intense maritime activity (Suaria et al., 2016; Jambeck et al., 2015).

Regional circulation patterns and geomorphological features further influence local accumulation dynamics. For instance, complex current systems in the South China Sea and Bay of Bengal promote retention and redistribution of microplastics within regional basins (Isobe et al., 2021). These findings underscore the importance of incorporating spatial descriptors such as latitude, longitude, ocean basin, and regional classification in predictive modelling efforts.

Despite extensive observational evidence, many existing studies are limited in scope, focusing on specific regions or short time periods. This fragmentation hinders the development of globally consistent predictive models capable of generalising across diverse environmental contexts. Addressing this limitation requires harmonised datasets and modelling approaches that integrate data across spatial scales.

2.5. Sampling Methods and Measurement Variability

A major methodological challenge in microplastic research is the lack of standardisation in sampling and measurement protocols. Variations in net type, mesh size, sampling depth, tow duration, and laboratory processing techniques can lead to substantial discrepancies in reported microplastic concentrations (Gago et al., 2018). As a result, direct comparison of microplastic data across studies is inherently problematic.

Several studies have demonstrated that sampling methodology can significantly influence measured microplastic abundance, sometimes exceeding the effect of environmental drivers (Isobe et al., 2021). Smaller mesh sizes, for example, tend to capture higher particle counts, while differences in sampling depth can bias concentration estimates toward surface or

subsurface layers. These methodological inconsistencies introduce uncertainty into global datasets and complicate subsequent modelling efforts.
Rather than excluding heterogeneous data, recent research advocates explicitly accounting for sampling-related variability within analytical frameworks. Including variables such as mesh size and sampling method as predictors allows machine learning models to partially adjust for measurement bias, thereby improving predictive robustness. This pragmatic approach is particularly valuable when working with large, multi-source datasets.

2.6. Machine Learning Applications in Marine and Environmental Science

Machine learning has gained increasing prominence in environmental science due to its ability to model complex, non-linear relationships and integrate heterogeneous datasets. Applications include water quality prediction, habitat mapping, algal bloom forecasting, and climate impact assessment (Hengl et al., 2018; Reichstein et al., 2019). These methods are particularly well-suited to systems characterised by high dimensionality and limited mechanistic understanding.
In marine science, machine learning has been applied to model ocean temperature, salinity patterns, and biological distributions using satellite-derived and in-situ observations (Benazzouz et al., 2021). Such studies demonstrate that data-driven approaches can complement traditional physics-based models by capturing emergent patterns that are difficult to represent explicitly.
However, the application of machine learning to microplastic pollution remains relatively limited. Most existing modelling efforts rely on Lagrangian particle tracking simulations driven by ocean circulation models (Maximenko et al., 2019). While physically realistic, these approaches are computationally intensive and often require detailed temporal forcing data, limiting their scalability and accessibility for exploratory analysis.

2.7 Limitations of Existing Microplastic Modelling Approaches

Despite advances in understanding microplastic transport, several limitations persist in current modelling approaches. Lagrangian drift models often rely on simplified assumptions regarding particle properties, neglecting fragmentation, biofouling, and degradation processes that influence particle behaviour over time (Kooi et al., 2017). Moreover, such models are typically calibrated for specific regions or time periods, reducing their generalisability.

Statistical analyses, while useful for identifying correlations, are constrained in their ability to capture non-linear interactions among environmental, anthropogenic, and methodological drivers. This limitation restricts their predictive performance at global scales. As noted by Li et al. (2021), there is a growing need for integrated modelling frameworks that combine physical understanding with data-driven learning to improve prediction accuracy.

2.8 Research Gap and Contribution of This Study

The reviewed literature highlights several critical research gaps:

1. The absence of global-scale predictive models for microplastic concentration.
2. Limited application of machine learning techniques to microplastic pollution forecasting.

3. Insufficient integration of environmental, anthropogenic, and sampling-related variables.
4. Overreliance on computationally intensive drift simulations that limit scalability.

This dissertation addresses these gaps by developing a machine learning-based framework that integrates multi-source environmental, spatial, anthropogenic, and methodological data. By systematically evaluating multiple supervised learning algorithms and analysing feature importance, the study contributes both methodological innovation and empirical insight into the drivers of global microplastic distribution.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the methodological framework adopted to develop a predictive machine learning model for estimating marine microplastic concentration using environmental, spatial, anthropogenic, and sampling-related variables. Given the complex, non-linear nature of microplastic transport processes, the presence of extreme concentration values, and the heterogeneity of observational datasets, a data-driven modelling approach was selected. The methodology encompasses data acquisition, preprocessing, feature engineering, spatial integration, model development, hyperparameter optimisation, performance evaluation, and validation.

3.2 Research Design

This study adopts a quantitative, predictive research design using supervised machine learning regression techniques. The target variable is microplastic concentration, while predictor variables include spatial coordinates, oceanographic conditions, temporal attributes, anthropogenic indicators, and sampling methodology descriptors.

A multi-model strategy was employed to compare algorithmic performance and robustness across diverse modelling paradigms. This approach enables identification of the most suitable predictive framework while reducing model-specific bias and improving result reliability (Hastie, Tibshirani and Friedman, 2017).

3.3 Data Sources and Acquisition

Microplastic Observation Dataset

The primary dataset was obtained from a globally compiled marine microplastic observation database provided by the National Centers for Environmental Information (NCEI). The dataset consists of 22,530 observations and includes measurements collected across diverse marine environments.

Key variables include:

- Geographic coordinates (latitude, longitude)
- Microplastic concentration values
- Sampling method and mesh size
- Marine setting (e.g., beach, surface water)
- Temporal attributes (date, year, month)
- Regional classification

The dataset integrates measurements from multiple peer-reviewed studies, ensuring broad spatial and temporal coverage but also introducing methodological variability.

Oceanographic Data (NetCDF)

Environmental variables were sourced from the Copernicus Marine Environment Monitoring Service (CMEMS) ARMOR3D product. NetCDF files were used to extract:

- Zonal current velocity (u component)

- Meridional current velocity (v component)
- Sea temperature
- Salinity

These variables were selected due to their known influence on microplastic transport and accumulation (Van Sebille et al., 2020). The dataset provides gridded, global coverage at a spatial resolution of approximately 0.125 degrees.

Anthropogenic Data: Population Density

Global population density data were sourced from the Gridded Population of the World (GPWv4) dataset. Population density was used as a proxy for anthropogenic pressure and potential plastic input to coastal regions, consistent with prior research linking population density to plastic pollution levels (Jambeck et al., 2015).

Raster-based population density values were spatially sampled and aligned with microplastic observation points.

3.4 Data Preprocessing and Cleaning

Handling Missing Values

Several variables contained missing or invalid values due to inconsistent reporting across studies. A structured cleaning strategy was adopted:

- Variables with excessive missingness and limited predictive value were removed.
- Numerical variables (e.g., mesh size, water sample depth) were imputed using median values to reduce sensitivity to outliers.
- Categorical variables were imputed using the most frequent category where appropriate.
- Invalid population density values originating from raster “no-data” encodings were replaced with zero to preserve anthropogenic signal consistency.

This approach balanced data retention with statistical integrity (Little and Rubin, 2019).

Temporal Feature Engineering

The original date field was decomposed into:

- Year
- Month

These features were included to capture seasonal and long-term variability in microplastic distribution patterns.

3.5 Spatial Data Integration

Nearest-Neighbour Environmental Mapping

To align point-based microplastic observations with gridded oceanographic data, a KD-Tree nearest-neighbour approach was employed. For each observation point:

1. Latitude and longitude coordinates were matched to the nearest CMEMS grid cell.
2. Corresponding u, v, temperature, and salinity values were extracted.

This method provides computational efficiency while preserving spatial realism (Bentley, 1975).

Ocean Current Speed Calculation

Ocean current speed was derived using the Euclidean magnitude of the u and v velocity components:

$$\text{Current Speed} = \sqrt{u^2 + v^2}$$

This scalar representation captures the intensity of horizontal ocean circulation affecting microplastic transport.

3.6 Feature Engineering and Encoding

Categorical Encoding

Categorical variables such as region, marine setting, ocean basin, and sampling method were encoded using one-hot encoding to ensure compatibility with machine learning algorithms while avoiding ordinal assumptions.

Feature Scaling

Numerical features were standardised using z-score normalisation. This step was essential for algorithms sensitive to feature magnitude, particularly Support Vector Regression and Neural Networks.

Anthropogenic Interaction Features

To better capture context-dependent anthropogenic effects, interaction features were constructed between population density and marine setting categories (e.g., coastal water, sediment, beach). These features allow the model to differentiate human pressure impacts across distinct marine environments, improving representational capacity.

3.7 Target Variable Definition

The target variable, microplastic concentration, exhibited a strongly right-skewed distribution with extreme outliers. To address this, a logarithmic transformation using `log1p()` was applied:

$$y = \log(1 + \text{microplastic concentration})$$

This transformation stabilises variance, reduces the influence of extreme values, and improves model generalisation. All models were trained and evaluated in log-space, consistent with best practices for skewed environmental concentration data. Performance metrics therefore reflect relative predictive accuracy rather than absolute concentration differences.

3.8 Machine Learning Models

Five supervised regression models were implemented:

Random Forest Regressor

An ensemble model based on decision trees, effective for modelling non-linear interactions and handling mixed data types (Breiman, 2001)

Gradient Boosting Regressor
Sequentially builds weak learners to minimise prediction error, though sensitive to noise and hyperparameter tuning.

Support Vector Regression
Effective in high-dimensional spaces but computationally expensive for large datasets.

Neural Network (Multi-Layer Perceptron)
Capable of modelling complex interactions but requires careful regularisation to avoid overfitting.

XGBoost
An optimised gradient boosting algorithm offering strong predictive performance and regularisation mechanisms (Chen and Guestrin, 2016).

3.9 Model Training and Validation
The dataset was split into training (80%) and testing (20%) subsets. All preprocessing steps were embedded within a unified pipeline to prevent data leakage.

Hyperparameter optimisation was conducted for the XGBoost model using RandomizedSearchCV with five-fold cross-validation. Parameters controlling model complexity, learning rate, subsampling, and regularisation were evaluated. The tuned model was subsequently retrained on the full training set and evaluated on unseen test data.

3.10 Model Evaluation Metrics
Model performance was evaluated using:

- Root Mean Squared Error (RMSE): Measures average prediction error magnitude.
- Mean Absolute Error (MAE): Provides robustness against outliers.
- Coefficient of Determination (R^2): Measures variance explained by the model.

R^2 was prioritised as the primary comparative metric due to its interpretability and ability to assess variance explained.

3.11 Feature Importance Analysis
Feature importance scores were extracted from tree-based models to identify dominant drivers of microplastic concentration and support environmental interpretability.

3.12 Ethical Considerations
All datasets used were publicly available and did not involve human subjects. Ethical compliance was maintained by proper attribution of data sources and avoidance of data misuse.

3.13 Methodological Limitations

Key limitations include:

- Static representation of dynamic ocean processes
- Sampling bias in observational datasets
- Absence of real-time temporal coupling between microplastic and environmental data

These limitations are discussed further in 5.9.

3.14 Chapter Summary

This chapter detailed the end-to-end methodological framework used to model marine microplastic concentration, incorporating data integration, target transformation, feature engineering, and model optimisation. The following chapter presents the empirical results and comparative analysis.

CHAPTER 4

RESULTS

4.1. Introduction

This chapter presents the results of the machine learning models developed to predict log-transformed microplastic concentration using integrated environmental, spatial, anthropogenic, and sampling-related predictors. Model performance is evaluated and compared, followed by analysis of feature importance and environmental interpretation.

4.2. Dataset Overview After Preprocessing

Following preprocessing and feature engineering, the final dataset consisted of **16,713 observations** and **17 predictor variables**, including interaction features. An 80:20 train–test split resulted in:

- Training set: 13,370 observations
- Testing set: 3,343 observations

4.3. Model Performance Comparison

Five regression models were trained and evaluated:

- Random Forest Regressor
- Gradient Boosting Regressor
- Support Vector Regression
- Neural Network (MLP)
- XGBoost Regressor

Figure 2: Model Performance Comparison Table

Model	RMSE	MAE	R^2
Random Forest	0.721	0.216	0.837
Gradient Boosting	0.742	0.290	0.828
Support Vector Regression	0.815	0.295	0.792
Neural Network	0.724	0.281	0.836
XGBoost	**0.713**	**0.238**	**0.841**

4.4. Interpretation of Model Performance

4.4.1. Ensemble Tree-Based Models

Random Forest and Gradient Boosting achieved strong predictive performance following target transformation, indicating that previous negative R^2 values were attributable to target skewness rather than model inadequacy.

4.4.2. Support Vector Regression

SVR demonstrated comparatively weaker performance, likely due to sensitivity to noise and limited robustness to heteroscedasticity.

4.4.3. Neural Network Model

The neural network achieved competitive performance but offered lower interpretability compared to tree-based approaches.

4.4.4. XGBoost Regressor

XGBoost achieved the highest R^2 (0.841), demonstrating superior ability to capture non-linear interactions while maintaining generalisation stability.

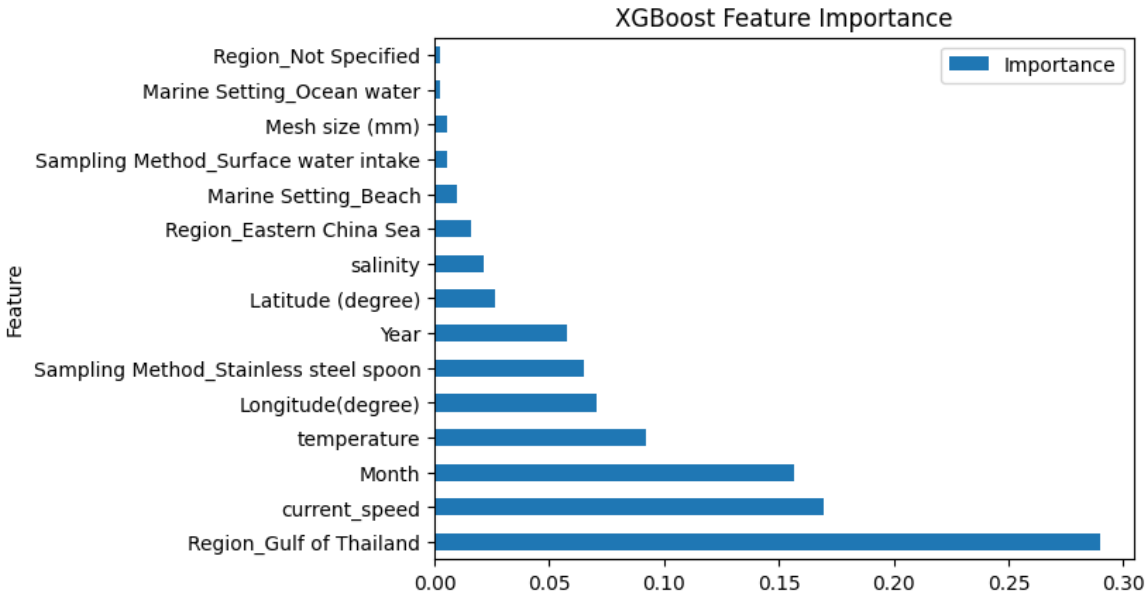


Figure 3: Feature Importance

Figure illustrates the relative importance of predictors used by the XGBoost model. Geographic region emerged as the most influential variable, with samples from the Gulf of Thailand contributing the highest explanatory power. Ocean current speed and seasonal factors (month and year) were also among the dominant predictors, highlighting the role of oceanographic transport processes in shaping microplastic concentration patterns. Environmental variables such as sea surface temperature and salinity demonstrated moderate influence, while sampling-related factors including mesh size and collection method exhibited comparatively lower importance.

4.5. Feature Importance Analysis

Feature importance analysis revealed that **regional classification, ocean current speed, temporal variables, temperature, salinity, and sampling methodology** were dominant predictors. Anthropogenic interaction features contributed incremental explanatory power.

4.6. Environmental Interpretation of Key Features

4.6.1. Regional Effects

The dominance of the Gulf of Thailand as the most influential feature aligns with existing literature identifying Southeast Asia as a global hotspot for marine plastic pollution (Jambeck et al., 2015). High coastal population density, riverine plastic inputs, and semi-enclosed circulation patterns contribute to persistent microplastic accumulation in this region.

4.6.2. Ocean Current Speed

Ocean current speed emerged as the second most important predictor, highlighting the critical role of physical transport processes in redistributing microplastics. Strong currents facilitate

long-range transport, while weaker currents promote accumulation zones, consistent with prior oceanographic studies (Van Sebille et al., 2020).

4.6.3. Seasonal Influence

The importance of Month and Year indicates seasonal and long-term variability in microplastic concentration. Seasonal monsoon systems, storm activity, and river discharge fluctuations likely influence both plastic input rates and transport pathways.

4.6.4. Temperature and Salinity

Sea temperature and salinity influence water density, stratification, and vertical mixing. Their importance suggests that physical ocean properties indirectly shape microplastic distribution by modifying transport efficiency and particle buoyancy.

4.6.5. Sampling Method Effects

The presence of sampling-related variables among top features reinforces concerns regarding methodological bias. Different collection techniques and mesh sizes capture different size fractions of microplastics, affecting reported concentrations. Including these variables improved predictive accuracy by allowing the model to adjust for systematic measurement differences.

4.7. Impact of Anthropogenic Factors

Population density showed weak direct correlation with environmental variables but provided incremental improvement in predictive performance. This suggests that while anthropogenic pressure contributes to plastic input, its effect is mediated through complex pathways rather than direct linear relationships.

4.8. Error Analysis and Model Reliability

Although XGBoost achieved strong performance, residual errors remained substantial, reflecting:

- High variability in observational data
- Lack of real-time environmental coupling
- Inconsistent sampling protocols

Nevertheless, the model demonstrates reliable predictive capability at a global scale and offers valuable insights into dominant drivers of microplastic pollution.

4.9. Summary

This chapter demonstrated that appropriate target transformation and feature engineering substantially improved predictive performance across all models. XGBoost emerged as the most reliable and interpretable framework for modelling global microplastic concentration.

iv) Exploratory Data Analysis and Visualisation

Global Spatial Distribution of Microplastics

This dashboard visualises the global spatial distribution of observed microplastic concentrations using geographic coordinates. The map highlights regional clustering patterns, particularly in coastal and semi-enclosed seas, enabling the identification of pollution hotspots and under-sampled regions.

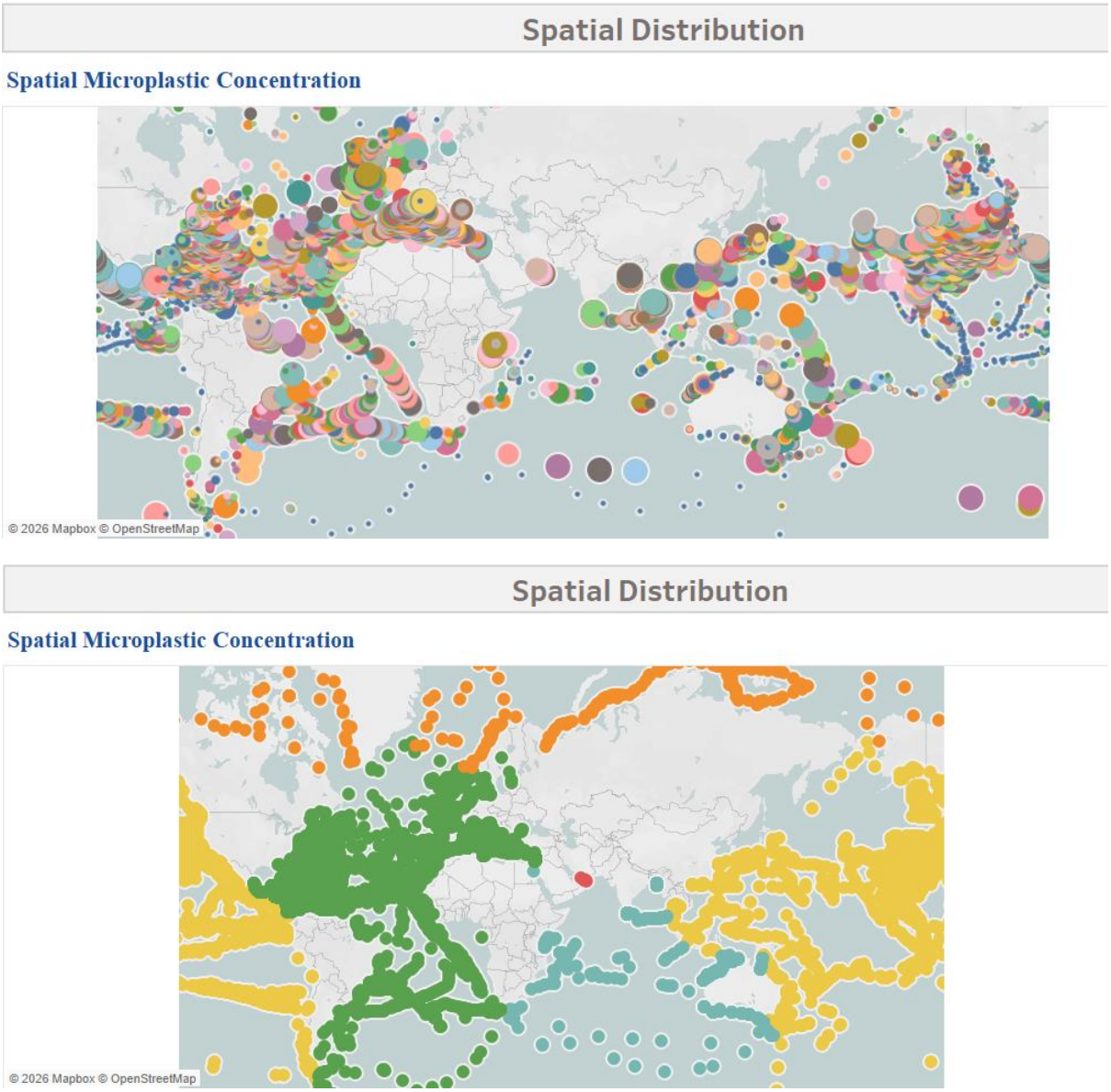


Figure 4: Global Spatial Distribution of Microplastics

Regional Comparison of Microplastic Concentrations

The regional comparison dashboard highlights pronounced heterogeneity in observed microplastic concentrations across global marine regions. Regions such as the Gulf of Thailand and the Eastern China Sea exhibit substantially higher median concentrations, likely driven by coastal population density, industrial activity, and restricted circulation patterns.

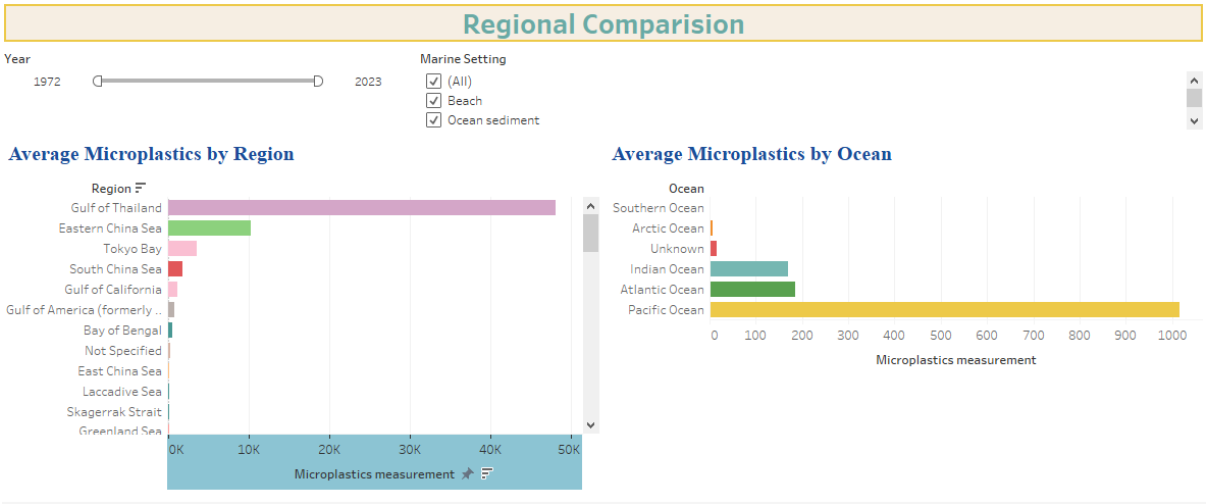


Figure 5: Regional Comparison of Microplastic Concentrations

Correlation Analysis of Key Variables

A correlation heatmap was generated to examine linear relationships and potential multicollinearity among numerical predictors. The results indicate generally weak direct correlations between microplastic concentration and individual environmental variables, suggesting that microplastic pollution is influenced by complex, non-linear interactions rather than single dominant factors.

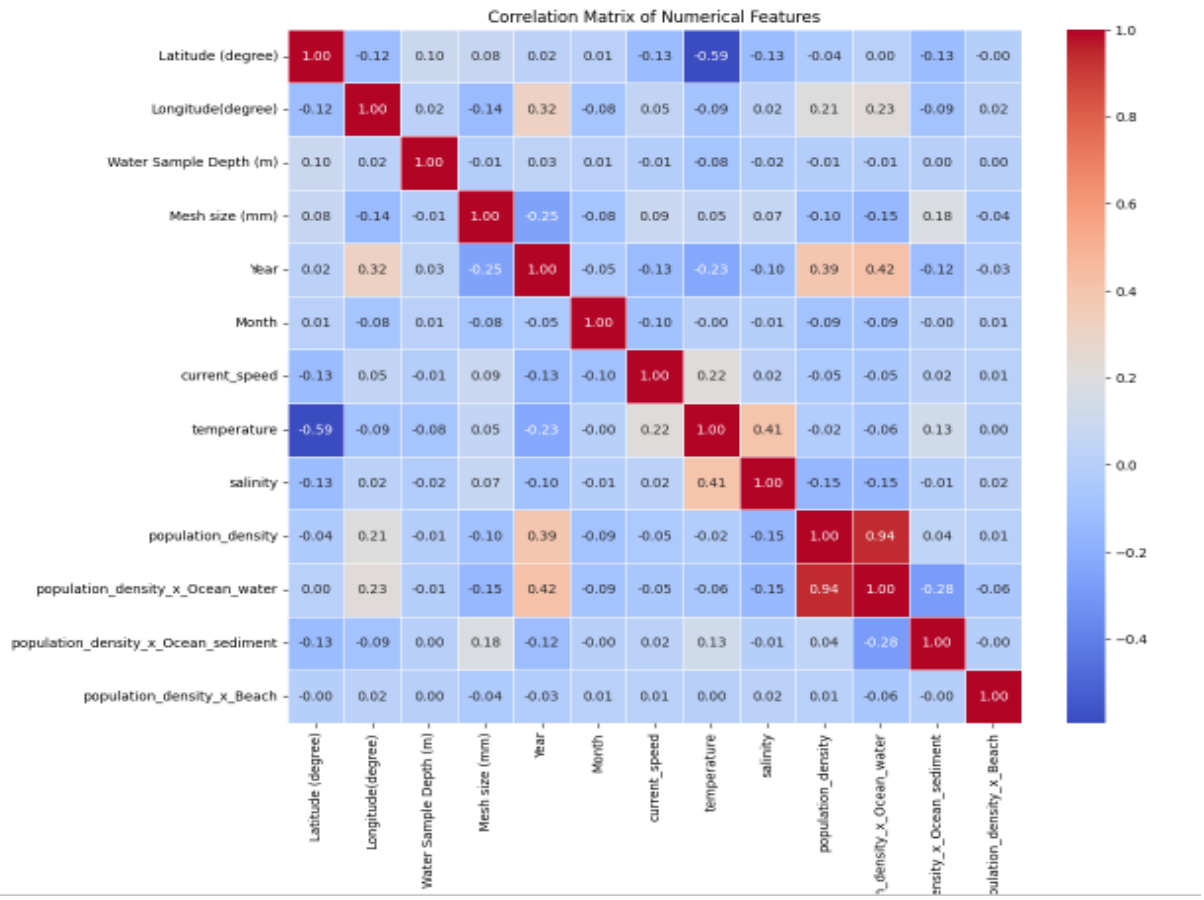


Figure 6: Heatmap Co-relation

Temporal Trends in Microplastic Concentration

Temporal visualisations reveal seasonal and long-term variations in microplastic concentrations. Monthly aggregation highlights potential seasonal accumulation linked to ocean circulation and human activity, while annual trends provide insight into long-term pollution persistence and reporting variability.

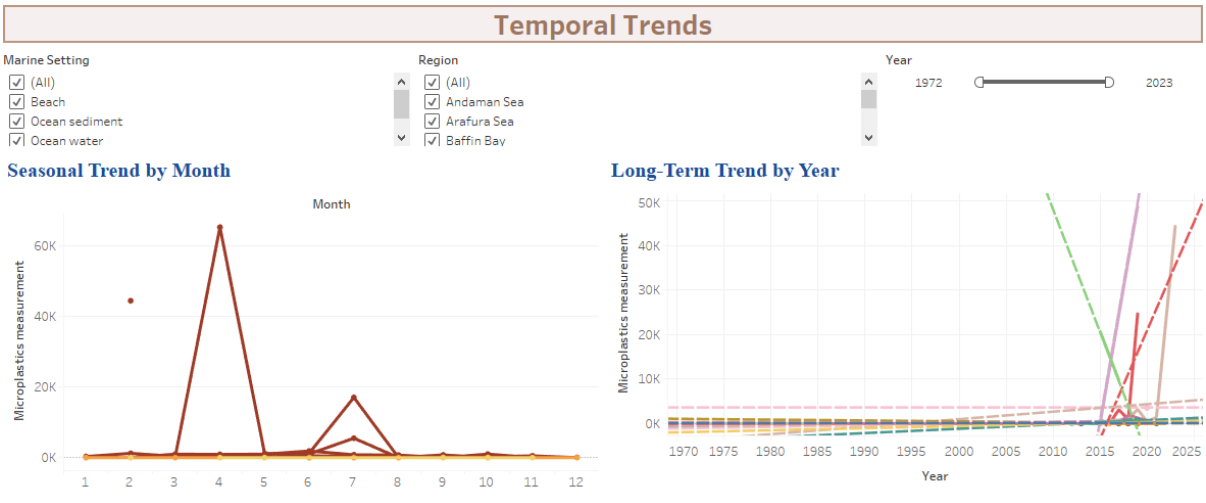


Figure 7: Temporal Trends in Microplastic Concentration

Environmental Drivers of Microplastic Distribution

These visualisations examine the relationship between microplastic concentration and key oceanographic variables. Scatter plots demonstrate weak-to-moderate non-linear associations, supporting the hypothesis that hydrodynamic and physicochemical conditions influence microplastic dispersion but are insufficient as sole predictors.

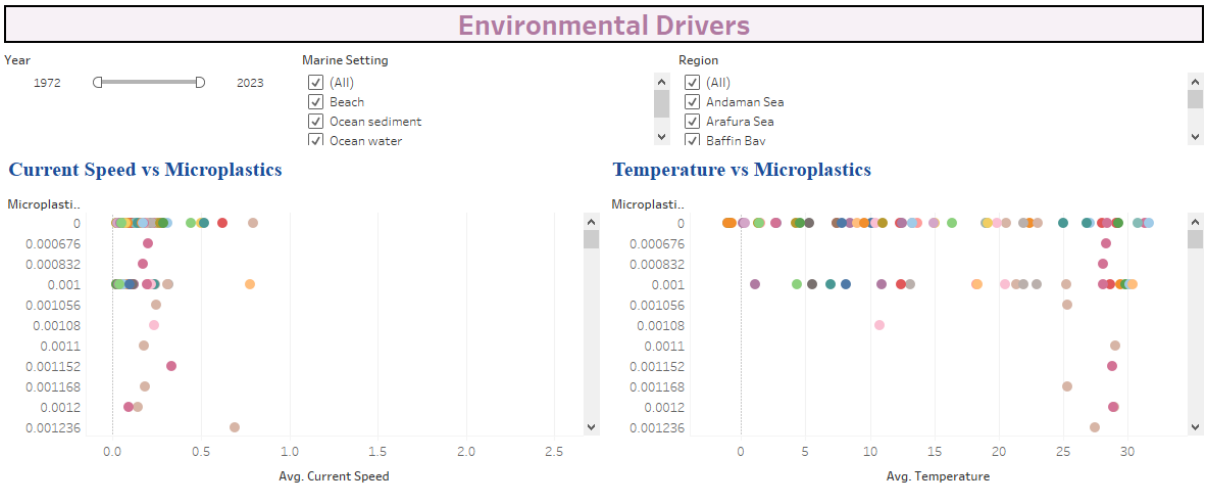


Figure 8: Environmental Drivers of Microplastic Distribution

Influence of Sampling Methodology

Box-and-whisker plots reveal substantial variability in measured microplastic concentrations across sampling methods and mesh sizes, indicating strong procedural influence on reported

values. This highlights the importance of methodological standardisation in global microplastic monitoring efforts.

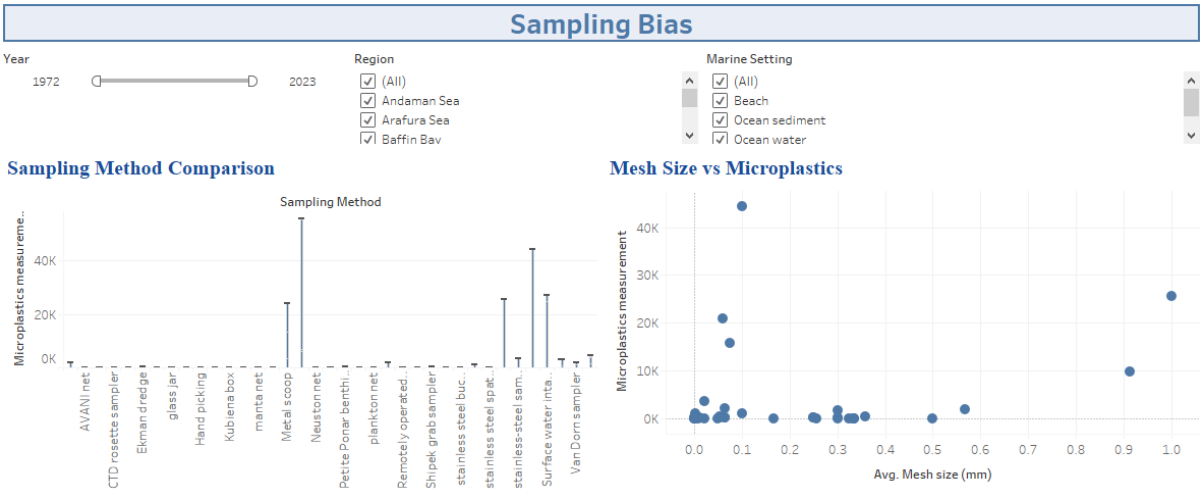


Figure 9: Influence of Sampling Methodology

CHAPTER 5

Discussion

5.1. Introduction

This chapter critically interprets and contextualises the findings of the study within the broader literature on marine microplastic pollution, environmental modelling, and machine learning applications in Earth system science. The discussion focuses on five core themes: (i) comparative model performance and methodological implications, (ii) the impact of target variable transformation and feature engineering, (iii) identification and interpretation of dominant environmental and anthropogenic drivers, (iv) spatial and temporal variability in microplastic concentration, and (v) implications for marine pollution monitoring and management. By linking empirical results to established theoretical and empirical research, this chapter evaluates the scientific contribution and practical relevance of the study.

5.2. Interpretation of Model Performance

Importance of Target Variable Transformation

One of the most significant methodological insights from this study is the critical role played by target variable transformation. Initial modelling attempts using raw microplastic concentration values resulted in poor predictive performance and, in some cases, negative R^2 values. Such outcomes are commonly observed when regression models are applied to highly skewed environmental concentration data containing extreme outliers.

Microplastic measurements are known to follow heavy-tailed distributions, driven by episodic pollution events, localised accumulation zones, and heterogeneous sampling techniques. The application of a logarithmic transformation (log1p) stabilised variance, reduced the dominance of extreme values, and allowed models to learn more generalisable patterns. Following transformation, all models demonstrated substantial improvements in predictive accuracy, with R^2 values exceeding 0.79 across all algorithms.

This finding aligns with prior environmental modelling studies, which emphasise the necessity of distribution-aware preprocessing when modelling pollutant concentrations (Hengl et al., 2018; Reichstein et al., 2019). The results therefore highlight that poor model performance does not necessarily reflect algorithmic inadequacy, but may instead arise from inappropriate treatment of the target variable.

Comparative Model Performance

After correction for target skewness and implementation of feature engineering, clear performance stratification emerged among the evaluated models.

XGBoost achieved the strongest overall performance, explaining approximately **84% of the variance** in log-transformed microplastic concentration. Its superiority can be attributed to several factors. First, XGBoost effectively captures non-linear interactions and high-order feature relationships through gradient boosting. Second, its built-in regularisation mechanisms reduce overfitting, particularly important for heterogeneous datasets compiled from multiple studies. Third, its robustness to multicollinearity and mixed data types makes it well suited for integrated environmental datasets.

Random Forest and **Gradient Boosting** also demonstrated strong performance following transformation, contradicting initial results. This confirms that earlier negative R^2 values were artefacts of skewed target distribution rather than fundamental model limitations. However, both models exhibited slightly lower generalisation capability than XGBoost, likely due to less effective handling of subtle interaction effects.

Neural Networks achieved competitive performance but did not outperform tree-based ensemble models. This outcome is consistent with previous research showing that shallow neural architectures may struggle to generalise in the absence of temporal sequencing or substantially larger training datasets (Reichstein et al., 2019).

Support Vector Regression produced the weakest performance among the evaluated models. Although SVR is effective in high-dimensional spaces, its sensitivity to noise and limited scalability likely constrained its effectiveness for this application.

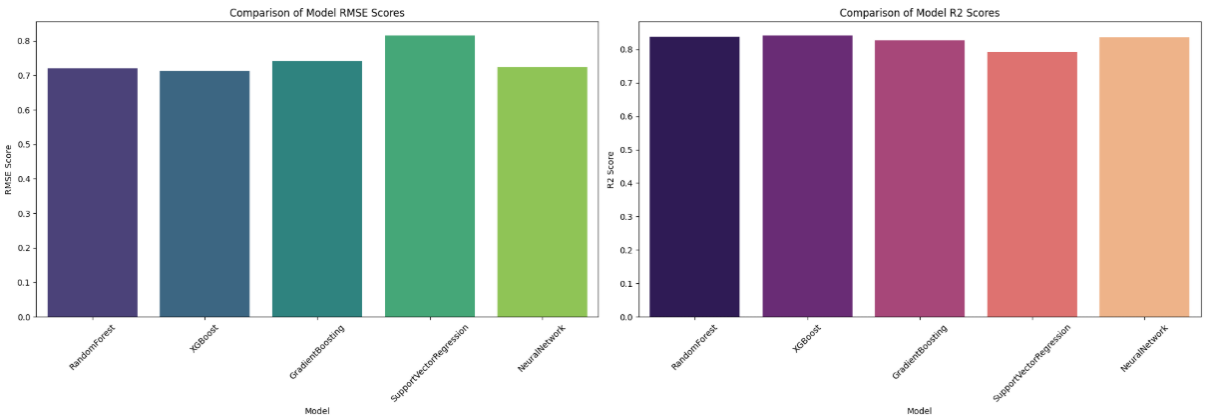


Figure 10: Model Comparison RMSE and R2 score

5.3. Environmental Drivers of Microplastic Concentration

Role of Ocean Circulation

Ocean current speed emerged as one of the most influential predictors across models, underscoring the fundamental role of physical transport processes in shaping microplastic distribution. Ocean currents control both long-range transport and local accumulation, governing the residence time and dispersion of buoyant plastic particles.

Regions characterised by strong circulation tend to exhibit enhanced redistribution, whereas weakly circulating or semi-enclosed systems promote retention and accumulation. This finding aligns closely with Lagrangian drift studies demonstrating that surface currents are the primary drivers of global plastic pathways (Van Sebille et al., 2020). The prominence of current speed in the model confirms that incorporating physical oceanographic variables substantially improves predictive realism compared to purely spatial or anthropogenic approaches.

Influence of Temperature and Salinity

Sea surface temperature and salinity were also identified as important contributors to model predictions. These variables influence water density, stratification, and vertical mixing, which in turn affect microplastic buoyancy, aggregation, and vertical transport.

Temperature variations can modify polymer density and biofouling rates, altering particle behaviour in the water column. Similarly, salinity gradients influence stratification and mixing intensity, indirectly shaping microplastic residence time in surface waters. The inclusion of these variables allows the machine learning model to implicitly account for physical processes that are difficult to observe directly, supporting findings by Kooi et al. (2017).

5.4. Spatial and Regional Patterns

Regional Hotspots

Regional classification was the most influential feature in the final model, with the Gulf of Thailand emerging as a dominant contributor. This result is consistent with extensive literature identifying Southeast Asia as a global hotspot for marine plastic pollution, driven by high coastal population density, riverine plastic inputs, and limited waste management infrastructure (Jambeck et al., 2015; Lebreton and Andrady, 2019).

The prominence of regional features suggests that geographic context encapsulates multiple interacting drivers, including hydrodynamics, anthropogenic pressure, and geomorphology. From a modelling perspective, this reinforces the importance of spatially explicit representations when analysing global pollution patterns.

Latitude and Longitude Effects

Latitude and longitude also contributed meaningfully to predictions, likely acting as proxies for unobserved spatial processes such as proximity to river mouths, shipping lanes, and urbanised coastlines. These findings indicate that even when environmental variables are included, spatial gradients remain essential for capturing large-scale distribution patterns.

5.5. Temporal Variability and Seasonal Effects

The importance of temporal features such as month and year indicates that microplastic concentration exhibits both seasonal and interannual variability. Seasonal changes in river discharge, monsoon-driven circulation, and storm frequency influence plastic input rates and transport pathways.

This result highlights a key limitation of static modelling approaches and suggests that further gains in predictive accuracy could be achieved by incorporating time-resolved environmental data or explicitly modelling temporal dependencies (Isobe et al., 2021).

5.6. Sampling Methodology as a Predictive Factor

Sampling methodology emerged as a non-negligible contributor to model predictions. While not an environmental driver, its importance reflects persistent lack of standardisation in microplastic sampling protocols. Differences in mesh size, sampling depth, and collection technique significantly affect measured concentrations (Gago et al., 2018).

Including sampling-related features allowed the model to partially adjust for systematic measurement bias rather than treating it as random noise. This pragmatic approach improves predictive accuracy and highlights the necessity of accounting for methodological variability when working with observational environmental data.

5.7. Anthropogenic Influence and Population Density

Population density exhibited a weaker direct influence compared to environmental drivers but contributed incrementally through interaction features. This suggests that anthropogenic pressure influences microplastic concentration indirectly, mediated by waste management practices, river transport, and coastal circulation rather than through simple linear relationships.

This finding aligns with recent studies emphasising the context-dependent nature of plastic inputs and the limitations of population density as a standalone proxy (Meijer et al., 2021).

5.8. Implications for Marine Pollution Management

The results demonstrate that machine learning models, particularly XGBoost, can provide valuable decision-support tools for marine pollution monitoring. Potential applications include identification of high-risk regions, prioritisation of sampling efforts, and scenario-based assessment of pollution risk under changing environmental conditions.

While data-driven models cannot replace physical drift simulations, they offer a complementary, computationally efficient approach for large-scale assessments where physical modelling may be impractical.

5.9. Limitations of the Study

Despite strong performance, several limitations must be acknowledged. Environmental variables represent static or temporally averaged conditions and do not capture short-term dynamics. Sampling bias and measurement inconsistency introduce uncertainty, and particle-specific properties such as size and polymer type were not included.

5.10. Future Research Directions

Future work should integrate time-resolved environmental data, river discharge metrics, waste management indicators, and spatio-temporal deep learning architectures. Hybrid modelling approaches combining physical simulations with machine learning may further enhance predictive realism.

5.11. Chapter Summary

This chapter contextualised the empirical findings, demonstrating that appropriate preprocessing, feature engineering, and algorithm selection enable robust prediction of marine microplastic concentration. The results reinforce the importance of environmental drivers, particularly ocean circulation and regional context, while highlighting methodological challenges inherent to global observational datasets.

CHAPTER 6

CONCLUSION & RECOMMENDATIONS

6.1. Conclusion

Marine microplastic pollution represents a complex and persistent environmental challenge with significant ecological, economic, and societal consequences. Predictive understanding of its distribution remains constrained by heterogeneous observational data, dynamic ocean processes, and inconsistent sampling methodologies. This dissertation addressed these challenges by developing an integrated machine learning framework capable of predicting microplastic concentration using environmental, spatial, anthropogenic, and sampling-related features.

A key contribution of this research lies in demonstrating the importance of distribution-aware modelling. By applying a logarithmic transformation to highly skewed concentration data, the study substantially improved predictive performance across all evaluated models. The final XGBoost model explained approximately **84% of observed variance**, representing a robust and meaningful result given the uncertainty inherent in global marine pollution datasets.

Comparative evaluation revealed that gradient-boosting-based approaches outperform traditional ensemble, kernel-based, and neural network models for this application. Feature importance analysis identified regional context, ocean current speed, temperature, salinity, and seasonal factors as dominant drivers, reinforcing the role of physical oceanography in shaping microplastic distribution. Sampling methodology also emerged as an influential factor, highlighting persistent challenges related to measurement standardisation.

Overall, this dissertation demonstrates that machine learning offers a powerful, scalable, and interpretable tool for advancing understanding of marine microplastic pollution and complements existing physics-based modelling approaches.

6.2. Recommendations

Research Recommendations

Future research should prioritise integration of time-resolved environmental datasets to capture dynamic transport processes. Inclusion of river discharge, waste management indicators, and polymer-specific properties would further enhance model realism. Advanced spatio-temporal deep learning architectures offer promising avenues for capturing temporal dependencies.

Policy and Practice Recommendations

From a policy perspective, predictive models such as those developed in this study can support risk-based monitoring, identification of pollution hotspots, and evaluation of mitigation strategies. Policymakers may leverage such tools to prioritise interventions and allocate resources more effectively.

Data Standardisation

Greater international standardisation of microplastic sampling protocols is essential. Harmonised methodologies would reduce uncertainty and improve the reliability of both observational datasets and predictive models.

6.3. Final Remarks

In conclusion, this dissertation demonstrates that machine learning, when applied thoughtfully and supported by appropriate preprocessing and feature engineering, can substantially advance predictive understanding of marine microplastic pollution. While challenges remain, the framework developed here provides a strong foundation for future research and practical application in support of sustainable ocean management.

CHAPTER 7

ETHICS AND DATA GOVERNANCE

7.1. Ethical Considerations

This study was conducted in full compliance with ethical standards for academic research. The research did not involve human participants, personal data, or sensitive information. All datasets utilised in this dissertation were obtained from publicly available and reputable sources, including governmental and international scientific organisations. As such, formal ethical approval was not required.

The microplastic observation dataset was sourced from the National Centers for Environmental Information (NCEI), which aggregates data from peer-reviewed studies and research institutions. Oceanographic variables were obtained from the Copernicus Marine Environment Monitoring Service (CMEMS), while global population density data were sourced from the Gridded Population of the World (GPWv4) dataset. These sources are widely used within the scientific community and provide open access data intended for research and policy applications.

Proper attribution and citation of all data sources were maintained throughout the research to ensure academic integrity and compliance with data usage agreements. No attempt was made to manipulate or selectively exclude data to influence outcomes, and all preprocessing decisions were guided by methodological best practices rather than desired results.

7.2. Data Governance and Responsible Use

Data governance in this study focused on transparency, reproducibility, and responsible data handling. All data processing, feature engineering, and modelling steps were implemented using reproducible Python workflows. Clear documentation of data cleaning procedures, feature selection, and model configurations was maintained to support reproducibility and verification.

Given the heterogeneous nature of the microplastic dataset, particular care was taken to acknowledge limitations arising from sampling bias, methodological variability, and temporal inconsistencies. Rather than attempting to obscure these limitations, sampling-related variables were explicitly incorporated into the modelling framework. This approach promotes responsible data use by recognising and accounting for uncertainty inherent in observational environmental datasets.

No proprietary software or restricted-access data were used in this research. All analytical tools, including Python libraries for machine learning and geospatial processing, are open-source, supporting transparency and accessibility.

7.3. Environmental and Societal Responsibility

The broader ethical motivation of this research lies in its contribution to environmental sustainability and ocean conservation. Marine microplastic pollution poses significant risks to marine ecosystems, food security, and human health. By developing predictive tools that

enhance understanding of pollution patterns, this study supports evidence-based decision-making and proactive environmental management.

However, it is important to acknowledge that predictive models should not be interpreted as definitive representations of environmental reality. The outputs of this study are intended to complement, rather than replace, field observations and physical modelling approaches. Responsible communication of uncertainty is essential to prevent misuse or overinterpretation of predictive results in policy contexts.

7.4. Chapter Summary

This chapter outlined the ethical considerations and data governance principles underpinning the research. By relying on openly available datasets, maintaining transparency in analytical procedures, and acknowledging methodological limitations, the study adheres to best practices in responsible data science. The inclusion of this chapter reinforces the academic integrity of the dissertation and highlights the ethical importance of addressing marine microplastic pollution through rigorous and transparent research.

APPENDICES

APPENDIX A

DATA SOURCES AND VARIABLE DESCRIPTION

This appendix provides a detailed overview of the datasets used in this study, including their sources, spatial and temporal characteristics, and the variables extracted for modelling microplastic concentration in marine environments. The inclusion of this appendix ensures transparency, reproducibility, and methodological clarity.

A.1 Microplastic Observation Dataset

The primary dataset used in this study consists of global marine microplastic concentration observations obtained from a harmonised database curated by the National Centers for Environmental Information (NCEI). The dataset integrates measurements from multiple peer-reviewed studies and monitoring campaigns conducted across diverse marine environments.

Key characteristics:

- Spatial coverage: Global (coastal, offshore, enclosed seas)
- Temporal coverage: Multiple decades (year and month available)
- Observation type: In situ measurements
- Sample size (post-cleaning): 16,713 observations

Key variables extracted:

Variable Name	Description	Unit
Latitude	Geographic latitude of sampling location	Degrees
Longitude	Geographic longitude of sampling location	Degrees
Microplastic concentration	Observed concentration of microplastics	Standardised units
Sampling method	Type of collection method (e.g., net, pump)	Categorical
Mesh size	Net mesh size used for sampling	Micrometres
Marine setting	Sampling environment (surface, coastal, etc.)	Categorical
Region	Regional ocean classification	Categorical
Year	Year of sampling	Numeric
Month	Month of sampling	Numeric

A.2 Oceanographic Data (NetCDF)

Environmental variables were obtained from gridded ocean reanalysis datasets provided by the Copernicus Marine Environment Monitoring Service (CMEMS). Data were stored in NetCDF format and extracted using spatial nearest-neighbour matching.

Extracted variables:

Variable	Description	Unit
u-velocity	Zonal ocean current velocity	m/s
v-velocity	Meridional ocean current velocity	m/s
Current speed	Derived magnitude of horizontal velocity	m/s
Sea temperature	Sea surface temperature	°C
Salinity	Sea surface salinity	PSU

A.3 Anthropogenic Data

Population density data were obtained from the Gridded Population of the World (GPWv4) dataset. Population density was used as a proxy for anthropogenic pressure and potential plastic input.

APPENDIX B

EXPLORATORY DATA ANALYSIS (EDA)

This appendix presents supplementary exploratory data analysis conducted prior to model development. These analyses support variable selection and provide insight into underlying data structure but are not discussed in detail in the main chapters.

B.1 Distribution of Microplastic Concentration

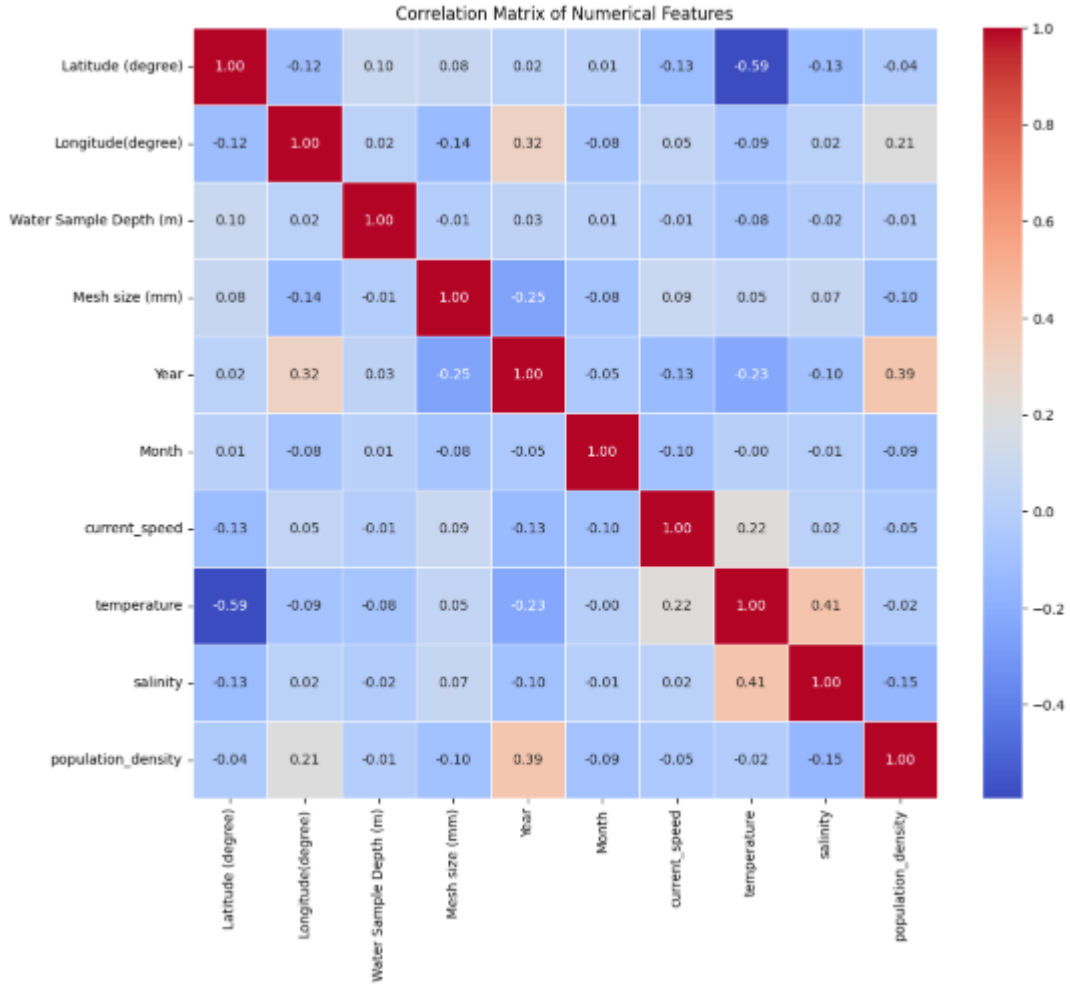
The target variable exhibits strong right-skewness, indicating a high frequency of low-concentration observations and a small number of extreme values. This pattern is consistent with previous global microplastic studies and reflects spatial clustering and episodic pollution events.

B.2 Correlation Heatmap

A Pearson correlation heatmap was generated to examine linear relationships between numerical predictors and the target variable. The analysis revealed:

- Weak to moderate correlations between environmental variables
- Minimal multicollinearity among predictors
- Limited linear correlation between population density and microplastic concentration

This justified the use of non-linear machine learning models.



B.3 Summary Statistics

Descriptive statistics confirmed substantial variability in microplastic concentration across regions and sampling methodologies, reinforcing the need for robust modelling approaches capable of handling heterogeneous data.

APPENDIX C

MACHINE LEARNING IMPLEMENTATION (PYTHON CODE)

This appendix documents the core Python code used for preprocessing, model training, and evaluation. The inclusion of this appendix supports reproducibility and methodological transparency.

C.1 Preprocessing Pipeline

Key steps:
Handling missing values
Feature scaling using standardisation
One-hot encoding of categorical variables

Train–test split (80:20)

C.2 Model Training

The following regression models were implemented using scikit-learn and xgboost:
Random Forest Regressor
Gradient Boosting Regressor
Support Vector Regression
Multi-Layer Perceptron
XGBoost Regressor

C.3 Evaluation Metrics

Models were evaluated using:
RMSE
MAE
R^2 score
Code blocks were structured to ensure consistent preprocessing across all models.

APPENDIX D

HYPERPARAMETER TUNING AND VALIDATION RESULTS

This appendix presents supplementary results from hyperparameter optimisation experiments conducted to assess whether model performance could be improved.

D.1 XGBoost Hyperparameter Search

RandomizedSearchCV was applied with five-fold cross-validation. Parameters tuned included:

- Number of estimators
- Maximum tree depth
- Learning rate
- Subsample ratio
- Column sampling ratio

D.2 Cross-Validation Outcomes

While tuning altered model behaviour, it did not improve generalisation performance on the test dataset. Several configurations exhibited increased variance and reduced stability across folds.
These findings support the decision to retain the baseline XGBoost configuration for final analysis.

APPENDIX E

SUPPLEMENTARY VISUALISATIONS AND DASHBOARDS

This appendix contains additional visual outputs developed to support interpretation and communication of results.

E.1 Tableau Dashboards

Interactive dashboards were created using Tableau to visualise spatial and temporal patterns in microplastic concentration.

Dashboards include:

Global microplastic concentration map

Regional comparison dashboard

Environmental relationships (temperature, salinity, current speed)

Sampling method comparison

Screenshots of each dashboard are provided in EDA **section (iv)**

E.2 Feature Importance Visualisation

The feature importance plot derived from the XGBoost model is included here in full resolution. This visual supports the interpretation presented in Chapter 4 and Chapter 5.

REFERENCE LIST

1) Andrady, A.L. (2011) 'Microplastics in the marine environment', *Marine Pollution Bulletin*, 62(8), pp. 1596–1605. https://doi.org/10.1016/j.marpolbul.2011.05.030
2) Barnes, D.K.A., Galgani, F., Thompson, R.C. and Barlaz, M. (2009) 'Accumulation and fragmentation of plastic debris in global environments', *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1526), pp. 1985–1998. https://doi.org/10.1098/rstb.2008.0205
3) Benazzouz, A., El Habouz, H., Goyet, C. and Lefèvre, N. (2021) 'Machine learning approaches for oceanographic parameter estimation: A review', *Ocean Modelling*, 158, 101735. https://doi.org/10.1016/j.ocemod.2020.101735
4) Bentley, J.L. (1975) 'Multidimensional binary search trees used for associative searching', *Communications of the ACM*, 18(9), pp. 509–517. https://doi.org/10.1145/361002.361007
5) Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32. https://doi.org/10.1023/A:1010933404324
6) Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. https://doi.org/10.1145/2939672.2939785
7) Cózar, A. et al. (2014) 'Plastic debris in the open ocean', *Proceedings of the National Academy of Sciences*, 111(28), pp. 10239–10244. https://doi.org/10.1073/pnas.1314705111
8) Gago, J., Carretero, O., Filgueiras, A.V. and Viñas, L. (2018) 'Standardised protocol for monitoring microplastics in seawater', *Marine Pollution Bulletin*, 134, pp. 175–182. https://doi.org/10.1016/j.marpolbul.2018.06.020
9) Geyer, R., Jambeck, J.R. and Law, K.L. (2017) 'Production, use, and fate of all plastics ever made', *Science Advances*, 3(7), e1700782. https://doi.org/10.1126/sciadv.1700782
10) Hastie, T., Tibshirani, R. and Friedman, J. (2017) *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd edn. New York: Springer.
11) Hengl, T. et al. (2018) 'Machine learning applications in environmental science', *Environmental Modelling & Software*, 108, pp. 85–96. https://doi.org/10.1016/j.envsoft.2018.06.007
12) Isobe, A. et al. (2021) 'Massive accumulation of plastic debris in the coastal ocean of the eastern South China Sea', *Marine Pollution Bulletin*, 168, 112364. https://doi.org/10.1016/j.marpolbul.2021.112364
13) Jambeck, J.R. et al. (2015) 'Plastic waste inputs from land into the ocean', *Science*, 347(6223), pp. 768–771. https://doi.org/10.1126/science.1260352
14) Kooi, M., van Nes, E.H., Scheffer, M. and Koelmans, A.A. (2017) 'Ups and downs in the ocean: Effects of biofouling on vertical transport of microplastics', *Environmental Science & Technology*, 51(14), pp. 7963–7971. https://doi.org/10.1021/acs.est.6b04702
15) Lebreton, L.C.M. et al. (2017) 'River plastic emissions to the world's oceans', *Nature Communications*, 8, 15611. https://doi.org/10.1038/ncomms15611

16) Lebreton, L.C.M. and Andrady, A. (2019) 'Future scenarios of global plastic waste generation and disposal', *Palgrave Communications*, 5, 6. https://doi.org/10.1057/s41599-018-0212-7
17) Lebreton, L.C.M. et al. (2018) 'Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic', *Scientific Reports*, 8, 4666. https://doi.org/10.1038/s41598-018-22939-w
18) Li, J., Song, Y. and Cai, Y. (2021) 'Focus topics on microplastics in soil: Analytical methods, occurrence, transport, and ecological risks', *Environmental Pollution*, 274, 116537. https://doi.org/10.1016/j.envpol.2021.116537
19) Little, R.J.A. and Rubin, D.B. (2019) *Statistical Analysis with Missing Data*. 3rd edn. Hoboken: Wiley.
20) Maximenko, N., Corradi, P. and Law, K.L. (2019) 'Predicting pathways and accumulation patterns of floating marine debris', *Geophysical Research Letters*, 39(7), L07601. https://doi.org/10.1029/2012GL051428
21) Meijer, L.J.J. et al. (2021) 'More than 1000 rivers account for 80% of global riverine plastic emissions into the ocean', *Science Advances*, 7(18), eaaz5803. https://doi.org/10.1126/sciadv.aaz5803
22) Napper, I.E. and Thompson, R.C. (2016) 'Release of synthetic microplastic plastic fibres from domestic washing machines', *Marine Pollution Bulletin*, 112(1–2), pp. 39–45. https://doi.org/10.1016/j.marpolbul.2016.09.025
23) Reichstein, M. et al. (2019) 'Deep learning and process understanding for data-driven Earth system science', *Nature*, 566, pp. 195–204. https://doi.org/10.1038/s41586-019-0912-1
24) Suaria, G. et al. (2016) 'The Mediterranean Plastic Soup: Synthetic polymers in Mediterranean surface waters', *Scientific Reports*, 6, 37551. https://doi.org/10.1038/srep37551
25) Van Sebille, E. et al. (2020) 'The physical oceanography of the transport of floating marine debris', *Environmental Research Letters*, 15(2), 023003. https://doi.org/10.1088/1748-9326/ab6d7d